

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-012: Global Dark Mode / Light Mode Toggle

Field	Details
Change Request ID	CR-012
Change Request Title	Global Dark Mode / Light Mode Toggle for the Student Management System
Student Name	AAM. Nusky
Student ID	JF/ICT/24/01
Implementation Status	Implementation report prepared; verify in the running application before marking complete
Date	9 October 2026

1. Change Request Summary

1.1 Business Requirement

The Student Management System requires a consistent theme control that allows users to switch between Light Mode and Dark Mode. The selected appearance should apply across the application rather than being maintained separately for each page. The selected theme should remain available after page navigation and browser refresh.

1.2 Existing Behaviour

The project already contains theme-related components and styles, including ThemeContext.jsx, ThemeToggle.jsx, global.css, landing.css, login.css, dashboard.css and studentDashboard.css. The theme context uses page/scope values such as home, login, admin and student, so the change consolidates the theme into one global state. Some component-specific CSS may still need additional dark-theme overrides.

1.3 Requested Change

- Provide one shared Light / Dark theme state for the entire frontend.
- Allow any existing theme toggle to change the global theme.
- Apply the selected theme to Home, Login, Admin Dashboard and Student Dashboard.
- Persist the selected theme in localStorage and restore it on reload.
- Use a document-level data-theme attribute so CSS can style all pages consistently.
- Style backgrounds, text, cards, forms, navigation, tables and modal components for both themes.
- Preserve existing routing, authentication, validation and API behavior.

2. Implementation

2.1 Implementation Summary

The proposed implementation centralizes theme state in ThemeContext.jsx. A single theme value ('light' or 'dark') is exposed through the ThemeProvider and useTheme hook. The context applies data-theme to the document root and stores the selection under the localStorage key student-management-theme. Existing ThemeToggle components can use the same global toggle, while CSS selectors provide theme-specific presentation.

2.2 Affected Files

File	Change	Purpose
frontend/src/context/ThemeContext.jsx	Modify	Replace per-page theme values with one global theme state, toggle function and persistence.
frontend/src/components/ThemeToggle.jsx	Review / minor update if needed	Ensure the button calls the global toggle and displays the appropriate label/icon.
frontend/src/main.jsx	Verify	Ensure the whole App is wrapped in ThemeProvider.
frontend/src/index.css	Modify	Add document-level theme rules and shared component overrides.
frontend/src/styles/landing.css	Review	Ensure landing page backgrounds, text and cards support both themes.
frontend/src/styles/login.css	Review	Ensure login form, labels, inputs, errors and card support both themes.
frontend/src/styles/dashboard.css	Review	Ensure admin dashboard tables, filters, buttons and modal styles support both

		themes.
frontend/src/styles/studentDashboard.css	Review	Ensure student dashboard profile and detail panels support both themes.
frontend/src/pages/LandingPage.jsx	Review	Remove duplicate toggle only if one toggle is preferred; keep existing layout and page logic.
frontend/src/pages/LoginPage.jsx	Verify	Confirm the page reads the shared theme and has no hard-coded light-only surfaces.
frontend/src/pages/DashboardPage.jsx	Verify	Confirm admin dashboard uses the global theme.
frontend/src/pages/StudentDashboardPage.jsx	Verify	Confirm student dashboard uses the global theme.
frontend/src/components/StudentFormModal.jsx	Review	Confirm modal fields and surfaces remain readable in dark mode.

2.3 Technical Changes

Global Theme State

- Initialize the theme from localStorage, defaulting to Light Mode if no valid saved value exists.
- Expose theme, toggleTheme and setTheme through ThemeContext.
- Use a functional state update to switch between 'light' and 'dark'.

Document-Level Theme Application

- Set document.documentElement's data-theme attribute to the current theme.
- Set the browser color-scheme property so native controls can follow the selected appearance.
- Use selectors such as html[data-theme="dark"] and html[data-theme="light"] in CSS.

Persistence and Compatibility

- Save the theme to localStorage under student-management-theme.
- Restore the saved theme when the application starts.
- Keep the existing useTheme API compatible where practical; review all existing calls before finalizing.
- Do not change authentication, API requests, database schemas or student data as part of this frontend change.

2.4 Implementation Logic

1. Load the saved theme from localStorage.
2. Initialize the shared ThemeProvider with the saved theme or Light Mode.
3. Render the application with the shared theme context.
4. When a user activates a toggle, switch the global theme state.
5. Apply the new data-theme attribute to the document root.
6. Update theme-specific CSS for all visible components.
7. Persist the selected theme so it survives a refresh.
8. Check each page and correct any component that remains in the wrong theme.

2.5 Error and Data Handling

- Theme storage is a UI preference only and does not contain account credentials or student records.
- If localStorage is unavailable, the application should safely fall back to Light Mode.
- Changing the theme must not submit forms, clear form values or alter authentication state.
- Existing application routes, API data and validation logic should remain unchanged.
- CSS overrides must preserve readable text contrast, focus indicators and input borders.

3. Testing & Verification

3.1 Testing Approach

This report defines the manual verification plan. Test results should be recorded only after the updated application has been run and the cases below have been executed.

- Functional testing of the theme toggle.
- Cross-page consistency testing.
- Persistence testing after browser refresh.
- UI testing for forms, cards, tables, navigation and modal dialogs.
- Regression testing for login, routing, validation and API-based operations.
- Accessibility checks for text contrast, focus visibility and keyboard operation.

Environment: Development (to be confirmed during testing)

Browser: Chrome / Edge / Firefox (record the browsers actually tested)

Test Roles: Administrator / Student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the application with no saved theme	Application opens in Light Mode by default.	To be recorded	Pending
TC-02	Activate Dark Mode toggle	All global surfaces switch to dark styling.	To be recorded	Pending
TC-03	Navigate to Login page	Login page uses the same selected theme.	To be recorded	Pending
TC-04	Open Admin Dashboard	Navigation, cards, filters, tables and forms follow the theme.	To be recorded	Pending
TC-05	Open Student Dashboard	Profile and detail panels follow the theme.	To be recorded	Pending
TC-06	Open Student Form modal	Modal, labels, inputs and buttons remain readable.	To be recorded	Pending
TC-07	Switch back to Light Mode	All pages return to light styling.	To be recorded	Pending
TC-08	Refresh the browser	The last selected theme is restored.	To be recorded	Pending
TC-09	Submit login / existing forms	Existing validation and submission flows still work.	To be recorded	Pending
TC-10	Keyboard and contrast review	Toggle is keyboard-operable and content remains legible.	To be recorded	Pending

3.3 Test Summary

Test Result	Count / Status
Total Test Cases	10
Passed	To be recorded after execution
Failed	To be recorded after execution
Pass Rate	Not calculated yet
Overall Test Result	Pending manual verification

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	A single toggle changes the global theme.	Toggle test across pages	Pending
AC-002	Theme is consistent on Home, Login, Admin and Student pages.	Navigate across all pages	Pending
AC-003	Theme preference survives browser refresh.	Refresh / localStorage test	Pending
AC-004	Cards, forms, tables and modal use appropriate colors.	UI review	Pending
AC-005	Light Mode remains readable and visually consistent.	Light theme UI review	Pending
AC-006	Existing login, validation, routes and API behavior remain intact.	Regression test	Pending
AC-007	Theme toggle does not submit forms or alter user data.	Interaction test	Pending

5. Evidence

Attach up to three screenshots after running the implementation:

- Evidence 01 – Home page in Dark Mode.
- Evidence 02 – Admin Dashboard in Dark Mode, including a table or modal.
- Evidence 03 – Light Mode and persistence after refreshing the browser.

Screenshots have not been inserted into this report; add genuine screenshots from the tested application.

6. Final Result

Item	Result
Requirement	Global theme behavior specified
Implementation	Proposed code changes documented; verify after applying
Testing	Pending manual execution
Acceptance Criteria	Pending verification
Existing Application Flows	Must be regression-tested
Final Status	Awaiting implementation verification

6.1 Final Summary

CR-017 defines a global Dark Mode / Light Mode enhancement for the Student Management System. The proposed change uses one shared theme state, a document-level data-theme attribute and localStorage persistence so the same theme can be applied across the application's pages. Existing theme-related components and CSS files are reused where possible. The implementation should be considered complete only after the affected pages have been updated, manual tests have passed and evidence screenshots have been attached.

6.2 Issues / Limitations

- Some components may contain hard-coded light colors and require page-specific dark overrides.
- Existing ThemeToggle calls and useTheme consumers must be checked for compatibility with the new global context.
- The report does not claim a passing test result because the updated application has not been verified in a running browser as part of report generation.

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Student ID: JF/ICT/24/01

Date: 9 October 2026

Signature: _____

~ END OF REPORT ~